

Game Development 1

What is game design?

Game Design

Game design is the process of creating and shaping the mechanics, systems, rules, and gameplay of a game.

Game design processes apply to board games, card games, dice games, casino games, role-playing games, sports, war games, or simulation games.

(https://en.wikipedia.org/wiki/Game_design)

Simple definitions?

Mechanics

How a game works for players.

A collection of rules and elements of play that define how a game operates for players.

Systems

How the game elements collectively interact, are organized, and communicate with each other.

The “whole” of the rules and mechanics.

Rules

Essential guidelines and procedures that define the games operation.

Gameplay

The interaction of the mechanics, systems, rules and player.

Mechanics

Mechanics

We are going to talk about mechanics as the actions the player takes in the game.

For example, jumping, shooting a projectile, playing a card, picking up a health potion would all be examples of mechanics.

Core Mechanic

I define this loosely as the most essential mechanic in the game.

It is the mechanic that all other mechanics rely on.

The core mechanic is the essential core action of the game. If you can't remove the mechanic without making the game non-existent.

Secondary Mechanics

I'm defining this as "Non-essential" mechanics. Systems such as health, powerups, skill trees, progression (such as leveling up) I'm considering as secondary mechanics.

A note on definitions

There's an active debate and some fuzziness around all these definitions, so take them with a grain of salt.

Systems

Systems

How the game elements collectively interact, are organized, and communicate with each other.

The “whole” of the rules and mechanics.

The “Whole”

Think about the game system as the interaction between rules and players, including elements that are not planned, defined or explicitly provided.

Emergent Elements

Systems often have a property of having emergent properties.

An emergent property is something that comes out of the collection interacting, but isn’t a specific defined part of the game.

Someone building a working computer chip within Minecraft would be an extreme example an emergent element.

It was never a designed feature, but something that “emerged” from the interaction of elements.

Additional reading

Thinking in Systems is a great book on recognizing and defining systems. The first few chapters should be available as a PDF online.

https://en.wikipedia.org/wiki/Thinking_In_Systems%3A_A_Primer

Metaphoric example of emergence: Water

Water Molecule

A single molecule of water is comprised of an Oxygen molecule and two Hydrogen molecules.

The “Whole”

A single water molecule will have different properties than a collection of water molecules (commonly called water).

Emergent Elements

The properties of water as a liquid are unique and different from the properties of the molecule, even though water emerges from the interaction of water molecules.

In fact the specific properties of a water molecule give water many specific properties (such as expanding when it freezes, the ease at which it mixes with other elements, the low melting temperature of ice, etc).

Back to games

Rules vs Systems

Like water, the most simple rules often lead to the most flexibility and ability to interact with many other elements.

The “Whole”

Emergent properties can be very difficult to design deliberately, and will often be discovered once players get their hands on your game.

Emergent Elements

Are often very interesting and exciting for players, and they often love exploring the game space looking for these emergent elements. This could be strategies, combinations of elements or interactions of rules.

Keep things simple to allow for emergent elements to reveal themselves.

Test the game often and get lots of players trying it to find emergent elements early.

**Ok So what? How does
this relate to games?**

**This is all very
abstract, so how can
we move towards a
unified theory of game
design?**

My attempt:

Games as a series of decisions.

Bringing things to this level of abstraction allows us to zoom out and consider all games similarly.

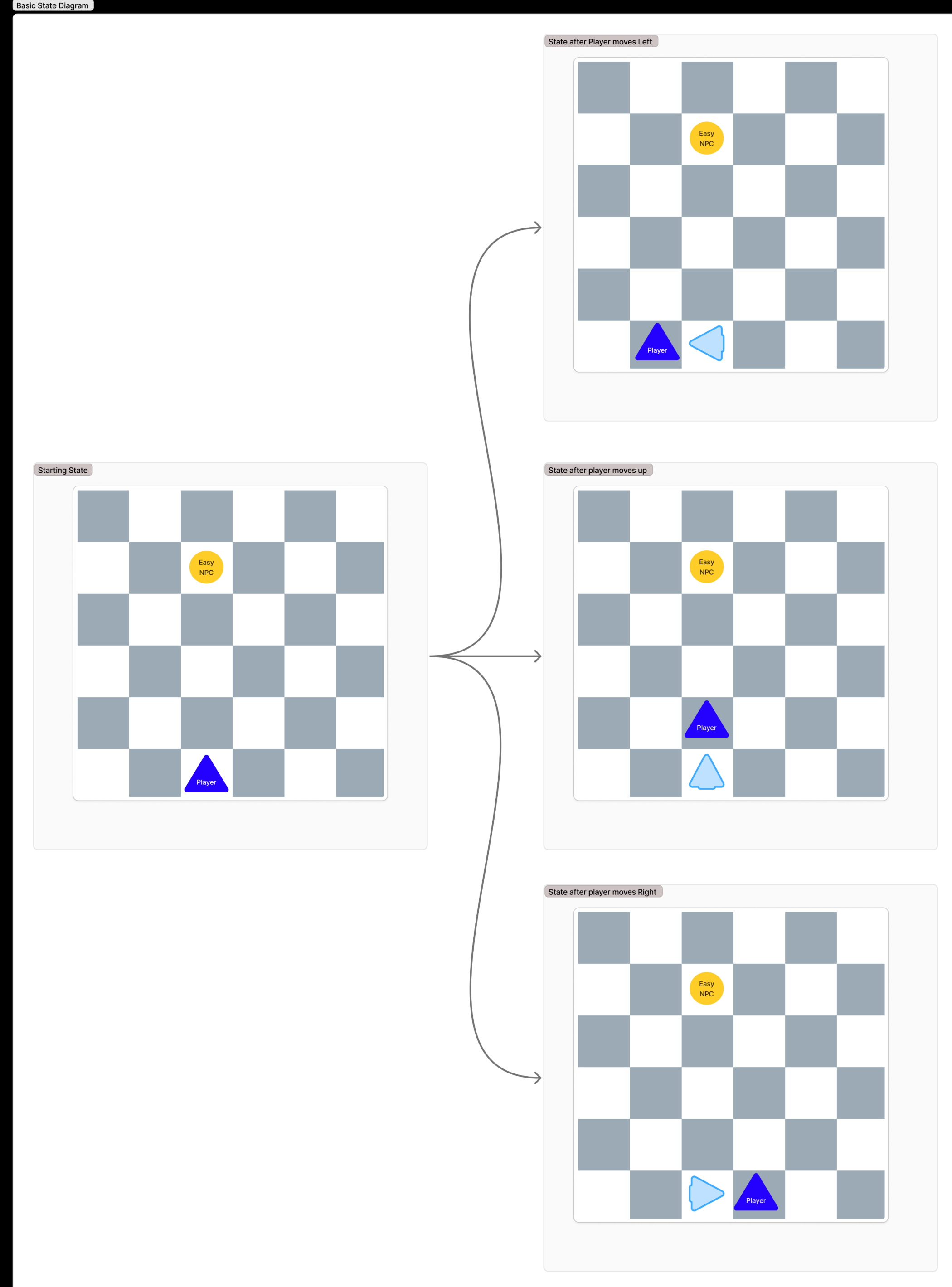
State Spaces

A useful way of thinking about games is the State Space model. This represents discrete states of the game as a sequence of options.

This is also sometimes called a Possibility Space

It is basically a map of which states will lead to other states.

It is very useful to think about games in this way as it puts it into a context of a series of decisions.



Ok Enough Theory Already!

Why is this useful?

This is providing us with a consistent framework to consider the game design of any game. By abstracting mechanics and the complexity into a common Graph framework, we have created a vehicle to think about ALL games consistent.

This is also very useful later when we get into behaviour trees and other AI topics....

How do we use this?

Think about the decisions you make in a game, and the order you make them.

You can even use this to think about actions in a real time game (really a series of very small timesteps or discrete steps still) by considering the moves abstractly (e.g. A move of a particular distance or time step.)

This lets you simplify otherwise overwhelmingly complex games.

Problems and limitations

As you are probably also aware, games are very complex, and drawing out a complete graph of a game will quickly become cumbersome.

Still, this is a useful mental framework for thinking about games, and gives us some math tools for thinking about sequences of decisions.

Today's activity: Play a card game.

What? Really?

Yes, this is a game design class after all.

Form up with a group (which I will randomly assign) and play one of the available games.

Consider how the Rules are presented. Are they simple or complex? Clear or confusing?

How did different people interpret them?

What are the end states of the game?

Whats the catch?

While you are playing the game, take some notes on a piece of paper (freeing you from sitting at your computers).

What mechanics does the game include? What would you say is the Core mechanics? What would be considered a Secondary Mechanic?

Are there any Emergent properties? What are they? What effect do they have on the game?

What Decisions are you having to make as a player? When are they Interesting Decisions, and when are they not Interesting Decisions?

Problems and limitations

Start with a simple game. You could play a simple card game, or one of the card games on hand.

Be prepared to discuss what you found. We will be building our own card games next week, so it's important you go through the evaluation process.